

# On a fixed point formula of Navarro–Rizo

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## Abstract

Let  $G$  be a  $\pi$ -separable group with a Hall  $\pi$ -subgroup  $H$  of order  $n$ . For  $x \in H$  let  $\lambda(x)$  be the number of Hall  $\pi$ -subgroups of  $G$  containing  $x$ . We show that  $\prod_{d|n} \prod_{x \in H} \lambda(x^d)^{\frac{n}{d} \mu(d)} = 1$ , where  $\mu$  is the Möbius function. This generalizes fixed point formulas for coprime actions by Brauer, Wielandt and Navarro–Rizo. We further investigate an additive version of this formula.

**Keywords:** fixed points, coprime action, Sylow subgroups,  $p$ -solvable groups,  $\pi$ -separable groups

**AMS classification:** 20D10, 20D20

## 1 Introduction

Navarro and Rizo [5] proved the following fixed point equation related to formulas of Brauer and Wielandt.

**Theorem 1** (Navarro–Rizo). *Let  $P$  be a finite  $p$ -group acting on a  $p'$ -group  $N$ . Then*

$$|C_N(P)| = \left( \prod_{x \in P} \frac{|C_N(x)|}{|C_N(x^p)|^{1/p}} \right)^{\frac{p}{|P|(p-1)}}. \quad (1.1)$$

For a finite group  $G$  and  $x \in P \in \text{Syl}_p(G)$  let  $\lambda_G(x)$  be the number of Sylow  $p$ -subgroups of  $G$  containing  $x$ . In the situation of Theorem 1, the Sylow  $p$ -subgroups of  $G := N \rtimes P$  have the form  $nPn^{-1}$  for  $n \in N$ . Note that  $N_G(P) = C_N(P)P$ . Suppose that  $x \in P \cap nPn^{-1}$ . Then there exists  $y \in P$  such that  $x = nyn^{-1}$ . Since  $[n, y] = xy^{-1} \in P \cap N = 1$ , it follows that  $x = y$  and  $n \in C_N(x)$ . Hence,  $\lambda_G(x) = |C_N(x) : C_N(P)|$  for all  $x \in P$ . Now (1.1) turns into the more elegant formula:

$$\prod_{x \in P} \lambda_G(x^p) = \prod_{x \in P} \lambda_G(x)^p. \quad (1.2)$$

We show that this holds more generally for all  $p$ -solvable groups. In fact, our main theorem applies to  $\pi$ -separable groups, where  $\pi$  is any set of primes. Recall that a  $\pi$ -separable group  $G$  has a unique conjugacy class of Hall  $\pi$ -subgroups. For a  $\pi$ -element  $x \in G$  let  $\lambda_G(x)$  be the number of Hall  $\pi$ -subgroups of  $G$  containing  $x$ .

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**Theorem 2.** *Let  $G$  be a  $\pi$ -separable group with a Hall  $\pi$ -subgroup  $H$  of order  $n$ . Then*

$$\prod_{d|n} \left( \prod_{x \in H} \lambda_G(x^d)^{\frac{n}{d}} \right)^{\mu(d)} = 1, \quad (1.3)$$

where  $\mu$  is the Möbius function.

For  $\pi = \{p\}$ , (1.3) becomes (1.2). We will show in Proposition 4 that (1.3) holds for arbitrary groups whenever they have a cyclic Hall  $\pi$ -subgroup (by a result of Wielandt, the Hall  $\pi$ -subgroups are conjugate in this situation too, see [2, Satz III.5.8]). In general, the left hand side of (1.2) can be larger or smaller than the right hand side (consider  $G = A_5$  and  $G = \text{GL}(3, 2)$  for  $p = 2$ ). We did not find a non-solvable group fulfilling (1.2) for  $p = 2$ .

In the last section we obtain the following additive version of (1.3).

**Theorem 3.** *Let  $G$  be a finite group with a Hall subgroup  $H$  of order  $n$ . Then*

$$\frac{1}{n^2} \sum_{d|n} \mu(d) \sum_{h \in H} \lambda_G(h^d)^{\frac{n}{d}}$$

is a non-negative integer, which is zero if and only if  $1 \neq H \trianglelefteq G$ .

## 2 The proof of Theorem 2

In the first step we reduce Theorem 2 to  $\pi$ -nilpotent groups. Let  $\alpha(G)$  be the left hand side of (1.3). Since

$$\alpha(G) = \left( \prod_{d|n} \prod_{x \in H} \lambda_G(x^d)^{\mu(d)/d} \right)^n,$$

we may replace  $n$  by the product of the prime divisors of  $|H|$ . Suppose that  $p \in \pi$  does not divide  $n$ . Then  $\langle x^d \rangle = \langle x^{dp} \rangle$  for all  $x \in H$  and  $d | n$ . Since  $\lambda_G(x^d)$  only depends on  $\langle x^d \rangle$ , it follows that

$$\prod_{d|np} \prod_{x \in H} \lambda_G(x^d)^{\frac{np}{d} \mu(d)} = \prod_{d|n} \prod_{x \in H} \lambda_G(x^d)^{\frac{np}{d} \mu(d)} \lambda_G(x^d)^{-\frac{n}{d} \mu(d)} = \alpha(G)^{p-1}.$$

Hence, we can assume that  $n = \prod_{p \in \pi} p$ .

Suppose that  $N := O_\pi(G) \neq 1$ . Since  $N$  lies in every Hall  $\pi$ -subgroup of  $G$ , we have  $\lambda_G(xy) = \lambda_G(x)$  for all  $x \in H$  and  $y \in N$ . Moreover,  $\lambda_G(x) = \lambda_{G/N}(xN)$ . By induction on  $|G|$ , we obtain

$$\alpha(G) = \prod_{x \in H} \prod_{d|n} \lambda_G(x^d)^{\frac{n}{d} \mu(d)} = \prod_{xN \in H/N} \left( \prod_{d|n} \lambda_{G/N}(x^d N)^{\frac{n}{d} \mu(d)} \right)^{|N|} = \alpha(G/N)^{|N|} = 1.$$

Thus, we may assume that  $O_\pi(G) = 1$ . Then  $N := O_{\pi'}(G) \neq 1$ . By the argument from the introduction, we have  $\lambda_{HN}(x) = |C_N(x) : C_N(H)|$  for  $x \in H$ . If  $x$  lies in another Hall  $\pi$ -subgroup  $K$ , then

$$\lambda_{HN}(x) = |C_N(x) : C_N(H)| = |C_N(x) : C_N(K)| = \lambda_{KN}(x),$$

because  $H$  and  $K$  are conjugate in  $G$ . It follows that  $\lambda_G(x) = \lambda_{HN}(x) \lambda_{G/N}(xN)$ . Hence, by the same argument as before,  $\alpha(G) = \alpha(HN) \alpha(G/N)$ . By induction on  $|G|$ , we may assume that  $G = HN$ , i. e.  $G$  is a  $\pi$ -nilpotent group.

Next, we reduce to the case where  $H$  is cyclic. Under this assumption, the result holds for arbitrary groups.

**Proposition 4.** *Let  $G$  be a group with a cyclic Hall subgroup  $H$ . Then (1.3) holds.*

*Proof.* For every generator  $h$  of  $H$  we have  $\lambda_G(h) = 1$ . Now let  $h \in H$  be of order  $\frac{n}{e} < n$ . Then for every  $d \mid e$  there exist exactly  $d$  elements  $x \in H$  such that  $x^d = h$ . Hence, the exponent of  $\lambda_G(h)$  in (1.3) is

$$\sum_{d \mid e} d \frac{n}{d} \mu(d) = n \sum_{d \mid e} \mu(d) = 0. \quad \square$$

For the general case, let  $\mathcal{Z}$  be the set of cyclic subgroups of  $H$ . Instead of running over all elements of  $H$ , we run over  $Z \in \mathcal{Z}$  and then over  $z \in Z$ . In order to track multiplicity, we use the Möbius function  $\mu$  of the lattice  $\mathcal{Z}$ . Since the subgroups of a cyclic group of order  $d$  are in bijection to the divisors of  $d$ , we have  $\mu(Z, W) = \mu(|W/Z|)$  whenever  $Z \leq W$  and  $\mu(Z, W) = 0$  otherwise. We define  $f(Z) := \sum_{W \in \mathcal{Z}} \mu(Z, W)$ . Recall the inversion formula for Euler's totient function:

$$\varphi(n) = \sum_{d \mid n} \frac{n}{d} \mu(d). \quad (2.1)$$

For  $W \in \mathcal{Z}$ , let  $[W]$  be the set of generators of  $W$ . For any function  $\gamma : G \rightarrow \mathbb{Z}$ , we have

$$\prod_{w \in W} \gamma(w) = \prod_{Z \leq W} \prod_{z \in [Z]} \gamma(z).$$

By Möbius inversion, it follows that  $\prod_{w \in [W]} \gamma(w) = \prod_{Z \leq W} \prod_{z \in Z} \gamma(z)^{\mu(Z, W)}$  and

$$\prod_{x \in H} \gamma(x) = \prod_{W \in \mathcal{Z}} \prod_{w \in [W]} \gamma(w) = \prod_{W \in \mathcal{Z}} \prod_{Z \leq W} \prod_{z \in Z} \gamma(z)^{\mu(Z, W)} = \prod_{Z \in \mathcal{Z}} \left( \prod_{z \in Z} \gamma(z) \right)^{f(Z)}. \quad (2.2)$$

Counting the number of factors on both sides also reveals that

$$|H| = \sum_{Z \in \mathcal{Z}} |Z| f(Z). \quad (2.3)$$

Assuming  $G = NH$ , we have

$$\lambda_G(x) = |C_N(x) : C_N(H)| = |C_N(x) : C_N(Z)| |C_N(Z) : C_N(H)| = \lambda_{ZN}(x) |C_N(Z) : C_N(H)| \quad (2.4)$$

for  $x \in Z \in \mathcal{Z}$ . Now we can put everything together and apply Proposition 4:

$$\begin{aligned} \alpha(G) &\stackrel{(2.2)}{=} \prod_{Z \in \mathcal{Z}} \left( \prod_{z \in Z} \prod_{d \mid n} \lambda_G(z^d)^{\frac{n}{d} \mu(d)} \right)^{f(Z)} \stackrel{(2.4)}{=} \prod_{Z \in \mathcal{Z}} \left( \alpha(NZ) \prod_{d \mid n} |C_N(Z) : C_N(H)|^{|Z| \frac{n}{d} \mu(d)} \right)^{f(Z)} \\ &\stackrel{(2.1)}{=} \prod_{Z \in \mathcal{Z}} |C_N(Z) : C_N(H)|^{\varphi(n) |Z| f(Z)} \stackrel{(2.3)}{=} \left( |C_N(H)|^{-|H|} \prod_{Z \in \mathcal{Z}} |C_N(Z)|^{|Z| f(Z)} \right)^{\varphi(n)}. \end{aligned}$$

At this point, the claim follows from Wielandt's formula [8, Satz 2.3], which we prove for sake of self-containment.

**Theorem 5** (Wielandt). *Let  $H$  be a group acting coprimely on a group  $N$ . Then*

$$|C_N(H)|^{|H|} = \prod_{Z \in \mathcal{Z}} |C_N(Z)|^{|Z| f(Z)}.$$

*Proof.* Since Wielandt's paper is hard to follow (even for a German native speaker), we use some modern ingredients. We consider  $N$  as a  $H$ -set via conjugation. By a theorem of Hartley–Turull [1, Lemma 2.6.2] (see also [7, Satz 9.20]), there exist a direct product  $A$  of elementary abelian groups and an isomorphism of  $H$ -sets  $\varphi : N \rightarrow A$ , i. e.  $\varphi(n^h) = \varphi(n)^h$  for  $n \in N$  and  $h \in H$ . It follows that  $\varphi(C_N(Z)) = C_A(Z)$  for  $Z \in \mathcal{Z}$ . Thus, we may replace  $N$  by  $A$ . Then  $N$  decomposes into its (characteristic) Sylow subgroups  $N = N_1 \times \dots \times N_k$ . Since  $C_N(Z) = C_{N_1}(Z) \times \dots \times C_{N_k}(Z)$ , we may assume further that  $N = N_1$  is elementary abelian. Thus,  $N$  is an  $\mathbb{F}_p H$ -module for some prime  $p$  not dividing  $|H|$ . The corresponding Brauer character  $\chi : H \rightarrow \mathbb{C}$  can be regarded as an ordinary character since  $|H|$  is coprime to  $p$ . We further extend  $\chi$  to the complex group algebra  $\mathbb{C}H$ . For  $S \subseteq H$  let  $S^+ := \sum_{s \in S} s \in \mathbb{C}H$ . The additive version of (2.2) reads

$$H^+ = \sum_{Z \in \mathcal{Z}} f(Z) Z^+.$$

By the first orthogonality relation,  $\chi(Z^+) = |Z|[\chi_Z, 1_Z]$ , where  $[\chi_Z, 1_Z]$  is the multiplicity of the trivial character  $1_Z$  as a constituent of the restriction  $\chi_Z$ . On the other hand, we have  $|C_N(Z)| = p^{[\chi_Z, 1_Z]}$ . It follows that

$$|C_N(H)|^{|H|} = p^{|H|[\chi, 1_H]} = p^{\chi(H^+)} = p^{\sum_{Z \in \mathcal{Z}} |Z|[\chi_Z, 1_Z] f(Z)} = \prod_{Z \in \mathcal{Z}} |C_N(Z)|^{|Z| f(Z)}. \quad \square$$

The proof of Theorem 5 relies on the Feit–Thompson theorem to guarantee that  $H$  or  $N$  is solvable. In comparison, the proof of Theorem 1 does not require representation theory, but uses the fact that Sylow subgroups are nilpotent.

### 3 The proof of Theorem 3

A  $\pi$ -element  $x \in G$  lies in a Hall  $\pi$ -subgroup  $H$  if and only if  $x \in N_G(H)$ . Hence, the map  $\lambda_G : H \rightarrow \mathbb{Z}$  is the permutation character of the conjugation action of  $H$  on the set  $\text{Hall}_\pi(G)$  of all Hall  $\pi$ -subgroups of  $G$  (we do not assume that these subgroups are conjugate in  $G$ ). We use the following recipe to construct a related character.

**Theorem 6.** *Let  $\chi$  be a character of a finite group  $H$  and let  $\alpha$  be a character of a permutation group  $A \leq S_n$ . For  $a \in A$  let  $c_i(a)$  be the number of cycles of  $a$  of length  $i$ . Then the map  $\chi_\alpha : H \rightarrow \mathbb{C}$  with*

$$\chi_\alpha(h) = \frac{1}{|A|} \sum_{a \in A} \alpha(a) \prod_{i=1}^n \chi(h^i)^{c_i(a)}$$

*for  $h \in H$  is a character or the zero map.*

*Proof.* See [3, Theorem 7.7.7 and Eq. (7.7.9)]. □

We assume the notation of Theorem 3 and choose a cyclic subgroup  $A \leq S_n$  generated by a cycle of length  $n$ . Let  $\alpha \in \text{Irr}(A)$  be a faithful character. For  $B \leq A$  of order  $d$ ,  $\sum_{b \in [B]} \alpha(b)$  is the sum of the primitive roots of unity of order  $d$ . A simple Möbius inversion shows that this sum equals  $\mu(d)$ . Moreover, every  $b \in [B]$  is a product of  $n/d$  disjoint cycles of length  $d$ . Thus,  $c_d(b) = n/d$  and  $c_i(b) = 0$  for  $i \neq d$ .

We apply Theorem 6 with  $\chi = \lambda_G$ . For  $h \in H$  we compute

$$\chi_\alpha(h) = \frac{1}{n} \sum_{d|n} \sum_{\substack{a \in A \\ |\langle a \rangle| = d}} \alpha(a) \lambda_G(h^d)^{\frac{n}{d}} = \frac{1}{n} \sum_{d|n} \mu(d) \lambda_G(h^d)^{\frac{n}{d}}.$$

Taking the scalar product of  $\chi_\alpha$  and the trivial character of  $H$ , shows that

$$\beta_G(H) := \frac{1}{n^2} \sum_{h \in H} \sum_{d|n} \mu(d) \lambda_G(h^d)^{\frac{n}{d}}$$

is a non-negative integer. This confirms the first part of Theorem 3.

If  $H \trianglelefteq G$ , then  $\lambda_G(h) = 1$  for all  $h \in H$  and it follows that  $\beta_G(H) = 0$  unless  $H = 1$  (where  $\beta_G(H) = 1$ ). Now assume that  $H$  is not normal in  $G$ . In particular,  $n > 1$ . Let  $t := |\text{Hall}_\pi(G)|$ . If  $t = 2$ , then  $|G : N_G(H)| = 2$  and  $N_G(H) \trianglelefteq G$ . But this would imply that  $H^g = O_\pi(N_G(H)) = H$  for every  $g \in G$ . Hence,  $t \geq 3$ .

Next we investigate the contribution of  $d = 1$  to  $\beta_G(H)$ . Note that  $\lambda_G(h)^n$  is the number of fixed points of  $h$  on  $\text{Hall}_\pi(G)^n$ , acting diagonally. The number of orbits of  $H$  on  $\text{Hall}_\pi(G)$  is at least  $t^n/n$ . Using Burnside's lemma and the trivial estimate  $\lambda_G(h^d) \leq t$  for  $d \neq 1$  and  $h \in H$ , we obtain

$$n\beta_G(H) \geq \frac{1}{n} \sum_{h \in H} \lambda_G(h)^n - \frac{1}{n} \sum_{h \in H} \sum_{1 \neq d|n} t^{\frac{n}{d}} \geq \frac{t^n}{n} - \sum_{1 \neq d|n} t^{\frac{n}{d}}.$$

It suffices to show that

$$n \sum_{1 \neq d|n} t^{\frac{n}{d}} < t^n.$$

If  $n$  is a prime, this reduces to  $nt < t^n$  and we are done as  $t \geq 3$ . If  $n = 4$  and  $t = 3$ , the claim can be checked directly. In all other cases, one can verify that  $n \leq t^{\frac{n}{2}-1}$  and

$$n \sum_{1 \neq d|n} t^{\frac{n}{d}} \leq \sum_{k=0}^{n-1} t^k = \frac{t^n - 1}{t - 1} < t^n.$$

This finishes the proof.

We remark that the degree  $\chi_\alpha(1) = \frac{1}{n} \sum_{d|n} \mu(d) t^{n/d}$  has several interesting interpretations. For instance, if  $t$  is a prime power, then  $\chi_\alpha(1)$  is the number of irreducible polynomials of degree  $n$  over the finite field  $\mathbb{F}_t$  (see [6, Corollary 10.2.3]). In general,  $\chi_\alpha(1)$  is the rank of the  $n$ -th quotient of the lower central series of a free group of rank  $t$  (see [4, Theorem 5.11 and Corollary 5.12]).

We now give an interpretation of  $\beta_G(H)$  in a special case. Suppose that  $H$  is nilpotent with regular Sylow subgroups (for all primes). Then the  $d$ -powers form a subgroup  $H^d$ , and for every  $h \in H^d$  there exist exactly  $|H : H^d|$  elements  $x \in H$  with  $x^d = h$  (see [2, Hauptsatz III.10.5 and Satz III.10.6]). Burnside's lemma applied to  $H^d$  acting on  $\text{Hall}_\pi(G)^{n/d}$ , yields

$$\beta_G(H) = \frac{1}{n} \sum_{d|n} \mu(d) f_{n/d}(H^d),$$

where  $f_{n/d}(H^d)$  is the number of orbits of  $H^d$  on  $\text{Hall}_\pi(G)^{n/d}$ .

Finally, we comment on a curiosity. In order to produce similar quantities as  $\beta_G(H)$ , we may replace  $\lambda_G$  by any character of any finite group  $G$ . For instance, let  $\tau$  be the conjugation character of  $G = A_5$  on  $\text{Syl}_3(G)$ . We compute

[illegible]

Similar curious numbers arise from  $G \in \{S_5, \mathrm{PSL}(2, 9), \mathrm{PGL}(2, 9)\}$ . This can be explained by the presence of large powers of  $\tau(1) = 10$ .

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